

APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

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Statistics, parametric test

Exercise 16.3

In a random sample of size five the sum of the sample values equals 155, while the sum of their squares equals 4837. Find the mean and the variance of the sample.

$$s_n^{*2} = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{n-1} (\sum (x_i - a)^2 - n(\bar{x} - a)^2)$$

$$a = 0$$

$$s_n^{*2} = \frac{1}{n-1} (\sum x_i^2 - n\bar{x}^2)$$

$$\bar{x} = \frac{155}{5}$$

$$s_n^{*2} = \frac{1}{n-1} (\sum x_i^2 - n\bar{x}^2) = \frac{1}{4} \left(4837 - 5 \times \left(\frac{155}{5} \right)^2 \right) = 8$$

Exercise 16.4 (M1)

The volumes, in millilitres, of soft drink of a random sample of 10 bottles from a large batch were as follows:

499, 525, 498, 503, 501, 497, 493, 496, 500, 495.

Stating clearly your null and alternative hypotheses, test the claim that the mean volume of soft drink in bottles in the batch is 500 ml. Use a 95% confidence interval. Assume that, for the batch, volumes are normally distributed with a standard deviation of 3 ml.

z - test, since σ is given

$$\sigma = 3 \quad \bar{x} = 500.7$$

$$z = \frac{\bar{x} - m_0}{\sigma} \sqrt{n} \sim \mathcal{N}(0,1)$$

$$H_0: \mathbb{E}X = 500 = m_0$$

$$H_1: \mathbb{E}X \neq 500$$

$$z = \frac{500.7 - 500}{3} \sqrt{10} \approx 0.7379$$

Significance level: 5%

$$\alpha = 0.05 \quad \frac{\alpha}{2} = 0.025$$

$$|z| \approx 0.7379 \quad z_{1-\frac{\alpha}{2}} = 1.96$$

$|z| < z_{1-\frac{\alpha}{2}} \Rightarrow$ we accept H_0 with 95% significance level

$$\mathbb{E}X = 500$$

Exercise 16.4 (M1)

```
data=[499 525 498 503 501 497 493 496 500 495];  
[h,p] = ztest(data,500,3,'Alpha',0.05,'Tail','both')  
alpha = 0.05;  
x=-3:0.01:3;  
plot(x,normpdf(x))  
hold on  
z_alpha_2 = norminv(1-alpha/2);  
area(-z_alpha_2:0.01:z_alpha_2,normpdf(-z_alpha_2:0.01:z_alpha_2),'FaceColor','g')  
hold off
```

Exercise 16.8 (M2)

Experiments with a new variety of tomato are being conducted at an agricultural research station. The crop is grown under carefully controlled conditions on ten experimental plots and the yields, in kg per plot, are found to be as follows:

60.2, 63.4, 58.8, 63.6, 64.7, 62.5, 66.0, 59.1, 65.1, 62.0;

It is known that an established variety of tomato would have a mean yield of 60 kg per plot on the experimental plot. Test, at the 95% confidence interval, whether the new variety has a different mean yield, stating clearly your null and alternative hypotheses. State also the assumptions underlying your analysis.

Exercise 16.8

$$\bar{x} = 62.54$$

$$H_0: \mathbb{E}X = 60 = m_0$$

$$H_1: \mathbb{E}X \neq 60$$

$$\frac{\bar{x} - m_0}{s_n^*} \sqrt{n} \sim t_{n-1}$$

$$s_n^{*2} = \frac{1}{n-1} \sum_{k=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left(\sum x_i^2 - n\bar{x}^2 \right) \approx 6.29$$

$$s_n^* \approx 2.51$$

$$t = \frac{62.54 - 60}{2.51} \sqrt{10} \approx 3.2$$

$$\alpha = 0.05 \quad \frac{\alpha}{2} = 0.025$$

$$t_{1-\frac{\alpha}{2}} = 2.2622$$

$$|t| > t_{1-\frac{\alpha}{2}} \Rightarrow \text{we reject } H_0$$

Exercise 16.8 (M2)

```
data=[60.2, 63.4, 58.8, 63.6, 64.7, 62.5, 66.0, 59.1, 65.1, 62.0];  
[h,p] = ttest(data,60,'Alpha',0.05,'Tail','both')  
df = 9;  
alpha = 0.05;  
x=-3.3:0.01:3.3;  
axis([-3.3 3.3 0 0.5])  
hold on  
plot(x,tpdf(x,df))  
t_alpha_2 = tinv(1-alpha/2, df);  
area(-t_alpha_2:0.01:t_alpha_2,tpdf(-t_alpha_2:0.01:t_alpha_2,df),'FaceColor','g')  
hold off
```

Exam

```
load examgrades
```

```
x = grades(:,1);
```

%%%% Test the null hypothesis that the data comes from a normal distribution with mean $\mu = 65$ and standard deviation $\sigma = 10$, against the alternative that the mean is greater than 65.

```
[h,p] = ztest(x,65,10,'Tail','right')
```


Stock

```
load stockreturns
```

```
x = stocks(:,3);
```

%%%% Test the null hypothesis that the sample data are from a population with mean equal to zero at the 1% significance level.

```
[h,p,ci,stats] = ttest(x,0,'Alpha',0.01)
```

Patient

load patients

tabulate(Gender)

histogram(Height,10) %%% nbins = 10

mean(Height)

std(Height)

quantile(Weight,[0.025 0.25 0.50 0.75 0.975])

Literature

- ❑ Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- ❑ Sándor Baran: Probability theory and statistics
- ❑ Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- ❑ Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- ❑ Matlab examples:
<https://www.mathworks.com/help/examples.html>